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EXTENDED ABSTRACT

Simulating riverine heatwaves: a European reconstruction of river temperatures with an LSTM

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1. Introduction and objectives

Riverine heatwaves are events of anomalously high water temperature that persist over several days. They affect river ecosystems and can lead to declines in the reproduction of aquatic species, increased fish mortality, and a general decrease in water quality [van Hamel (2025)]. Over the past decades, an increase in the frequency and intensity of riverine heatwaves has been observed, likely linked to increases in air temperature and more frequent streamflow deficits [Tassone (2022)].

To better understand regional factors and climatic conditions leading to these events and to quantify long-term changes, long time-series of river temperatures are required over a large spatial domain. Existing observations, however, do not provide sufficient temporal and spatial coverage for large-scale and long-term analyses. Therefore, we here aim to generate a temporally and spatially coherent dataset over Europe by filling current data gaps.

2. Methods and data

Starting from a newly assembled European dataset of daily river temperature records called TempER (Temperature of European Rivers), we reconstruct continuous time-series at point locations to fill temporal and spatial data gaps. TempER contains water temperature data from over 4000 measurement stations in Europe, ranging from 1 to 72 years in length. For the reconstruction, we train a deep learning model (Long Short-Term Memory network, LSTM) forced by atmospheric reanalysis records and static variables related to landcover, topography, and geology. We evaluate the model with respect to reconstruction in time and in geographical regions without data coverage.

While average water temperature conditions are simulated well by the model, riverine heatwave characteristics – frequency, intensity, and duration – are less accurately reproduced. To address these limitations, we explore strategies to improve the simulation of riverine heatwaves. These include modifications to the loss function and the set of input variables, for example, integrating streamflow records whenever available and applying a composite loss function to incentivize deviations from air temperature.

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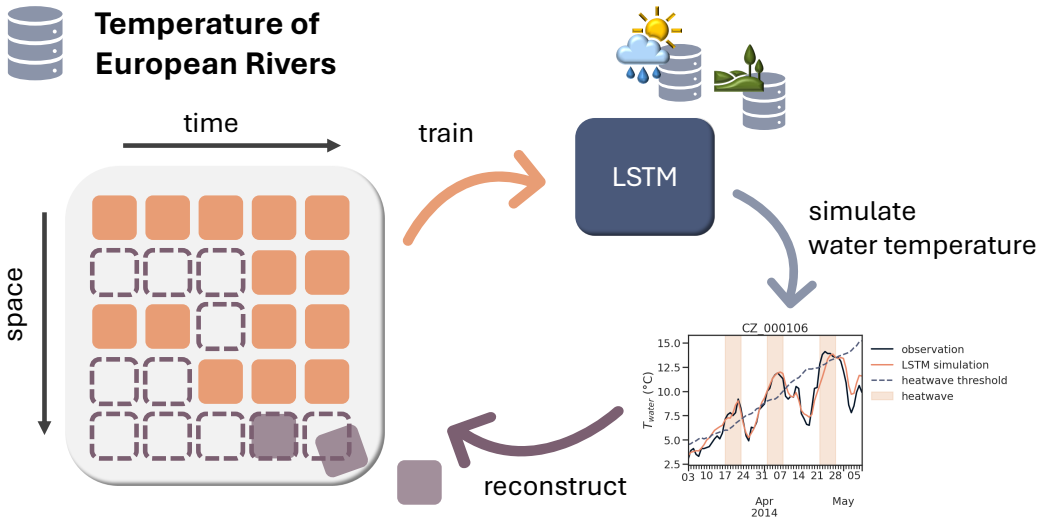

Figure 1. *Schematic overview of the presented work. Temporal and spatial data gaps of European records of river temperature (TempER dataset) are filled with an LSTM model to analyze trends and drivers of riverine heatwaves.*

3. Preliminary results and/or expected outcomes

Our work will result in a river temperature and riverine heatwave simulator to create a reconstruction dataset that is (1) spatially coherent, (2) spans a large time period and (3) covers the entire European continent. Such a dataset will allow robust trend analysis to ultimately improve our understanding of the hydro-meteorological processes that drive riverine heatwaves.

References

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